Inspect Before Modeling

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Inspect Before Modeling before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Inspect Before Modeling trace and record the first step you can explain without help.

Use W2: Interpret the audit counts for Dataset inspection and cards. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Inspect Before Modeling explanation against the supplied facts before

> discussing it.

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Inspect Before Modeling target to a trace and a justified conclusion.

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Inspect Before Modeling case and show the deciding evidence.

Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.

Name checks and a deletion risk.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 26

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 5 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	

Test record	My evidence
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Inspect Before Modeling adds to Responsible Plant Classifier and keep the evidence.

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Define the row and source

> Define the row and source. Apply the supplied Inspect Before Modeling facts and keep your reasoning.

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Interpret the audit counts

> Interpret the audit counts. Apply the supplied Inspect Before Modeling facts and keep your reasoning.

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Reject an unsupported correction

> Reject an unsupported correction. Apply the supplied Inspect Before Modeling facts and keep your reasoning.

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Inspect and document independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Inspect and document independently. Apply the supplied Inspect Before Modeling facts and keep your reasoning.

Rules supplied for this independent case

Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 State an evidence gap

State an evidence gap. Apply the supplied Inspect Before Modeling facts and keep your reasoning.

Someone **duplicate**s each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

WORKBOOK · 17 OF 26

Units conversions and data provenance — W1 Retain the source

> Use the supplied Inspect Before Modeling task to record your reasoning, evidence and next step.

Name four pieces of information to keep for Log A and its **conversion**.

My reasoning and evidence

Units conversions and data provenance — W2 Convert precisely

> Use the supplied Inspect Before Modeling task to record your reasoning, evidence and next step.

Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

My reasoning and evidence

Units conversions and data provenance — W3 Repair a comparison

> Use the supplied Inspect Before Modeling task to record your reasoning, evidence and next step.

A report claims 120 seconds exceeds 3 minutes because 120 > 3. Correct it.

My reasoning and evidence

WORKBOOK · 20 OF 26

Units conversions and data provenance — W4 Normalize independently

> Use the supplied Inspect Before Modeling task to record your reasoning, evidence and next step.

Create records for 240 seconds, 1.5 minutes and 60 seconds. Derive minutes while keeping raw data. State the values and their total.

My reasoning and evidence

WORKBOOK · 21 OF 26

Units conversions and data provenance — W5 Handle an unknown unit

> Use the supplied Inspect Before Modeling task to record your reasoning, evidence and next step.

A record contains value 8 and unit ticks with no **conversion** rule. Should you assume seconds?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Inspect Before Modeling

> Spot a duplicate, an impossible value and a valid boundary before fitting. Explain the labelled step before checking the changed condition.

Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.

Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/5

Inspect Before Modeling

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count rows

Five raw rows represent four distinct IDs; B is duplicated.



2. Check schema

C violates the nonnegative-height rule; D=0 is valid under this contract.



3. Flag before fitting

Flag duplicate B and invalid C. Do not count B twice or silently replace C.

CHANGE ONE THING

The teacher confirms B's second row is an accidental duplicate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Flag **duplicate** B and invalid C. Do not count B twice or silently replace C.

1. Flag before **fitting**

2. Count rows

3. Check **schema**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Flag before **fitting** Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. Read the picture in order. Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The teacher confirms B's second row is an accidental **duplicate**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Repair a missing building block

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

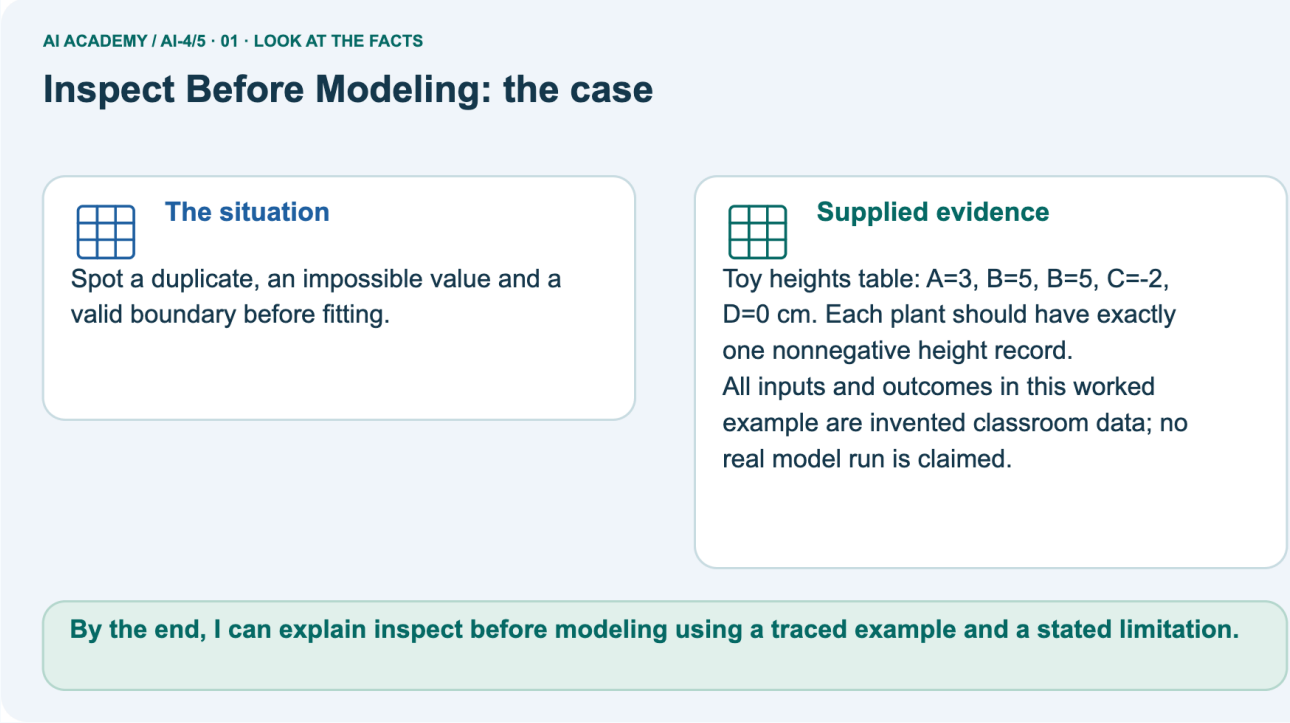
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Inspect Before Modeling picture from its labels, then record a relevant limit.

01. Inspect Before Modeling: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

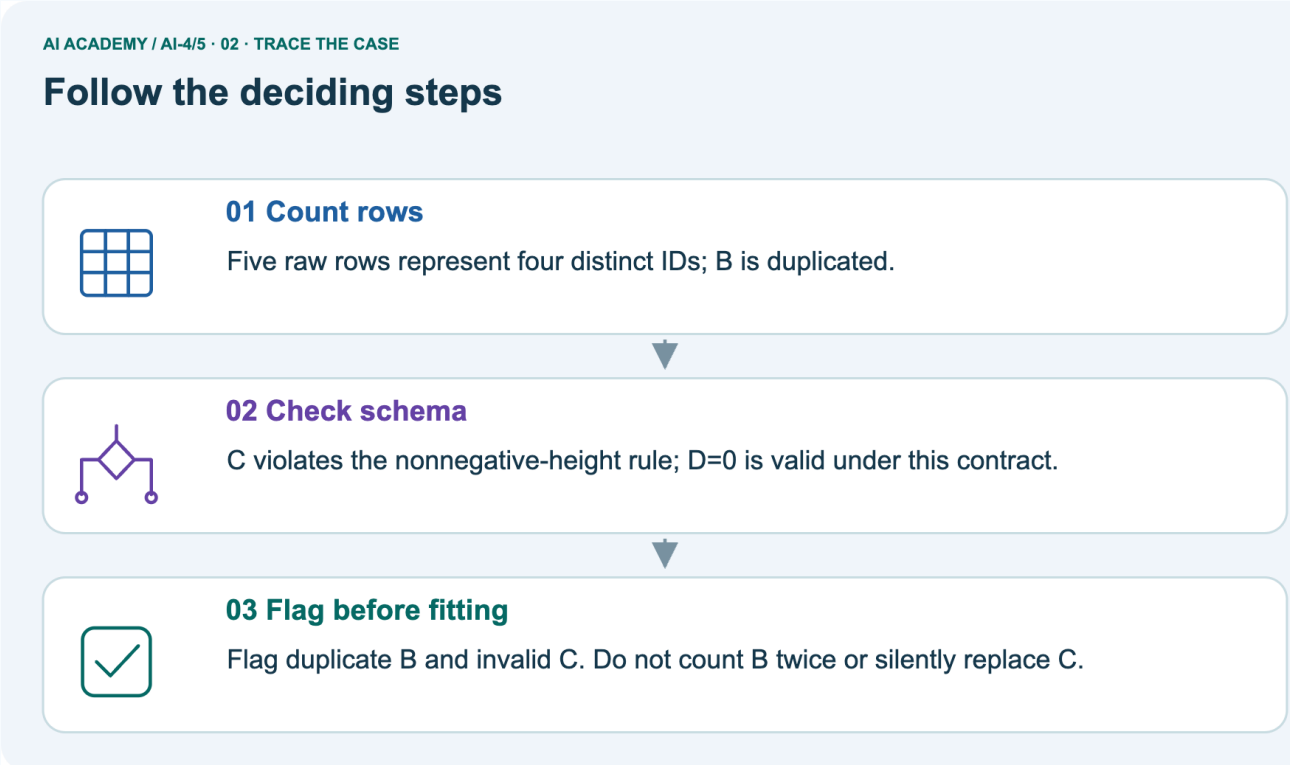
Spot a **duplicate**, an impossible value and a valid boundary before **fitting**. Supplied facts: Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Count rows: Five raw rows represent four distinct IDs; B is **duplicate**d. → Check **schema**: C violates the nonnegative-height rule; D=0 is valid under this contract. → Flag before **fitting**: Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action

AI ACADEMY / AI-4/5 · 03 · INSPECT THE LESSON MODEL

Turn an observation into a documented action

Finding	What it establishes	Next action
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

Follow the labelled picture with the matching explanation. Open the vector diagram

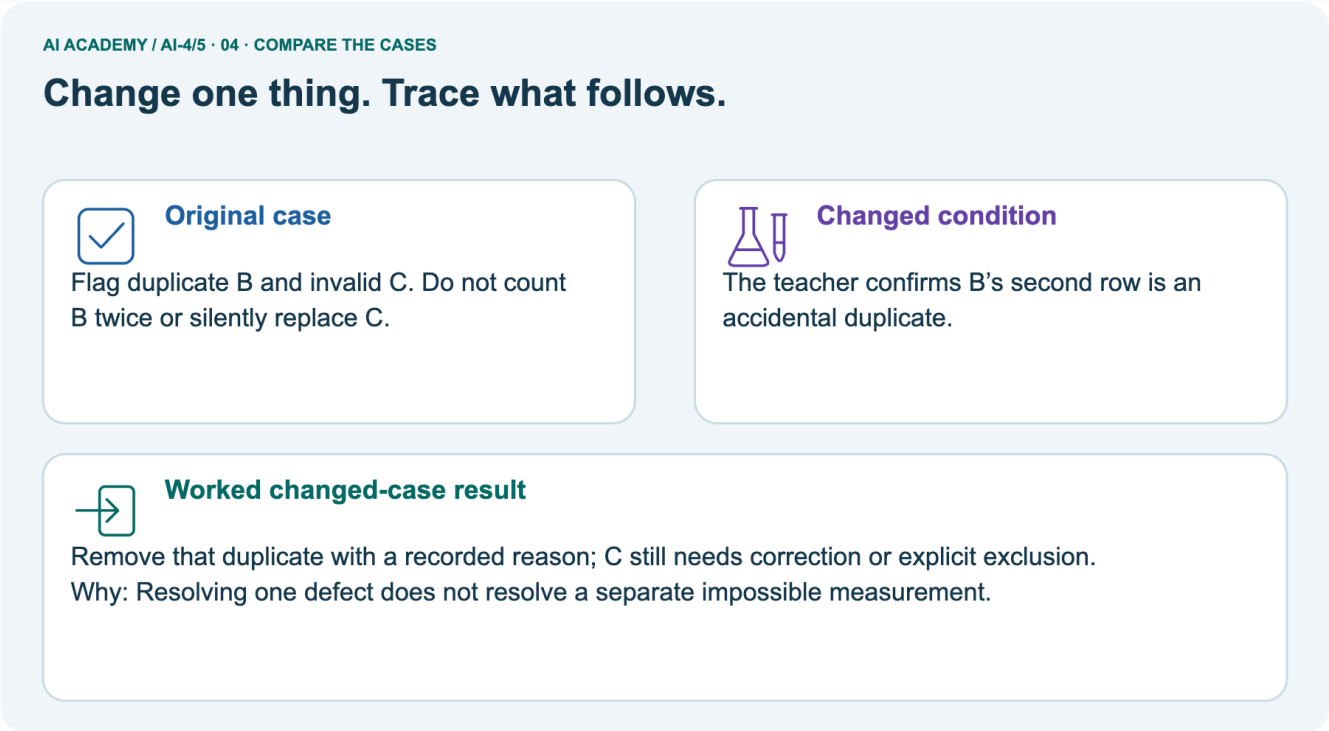
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

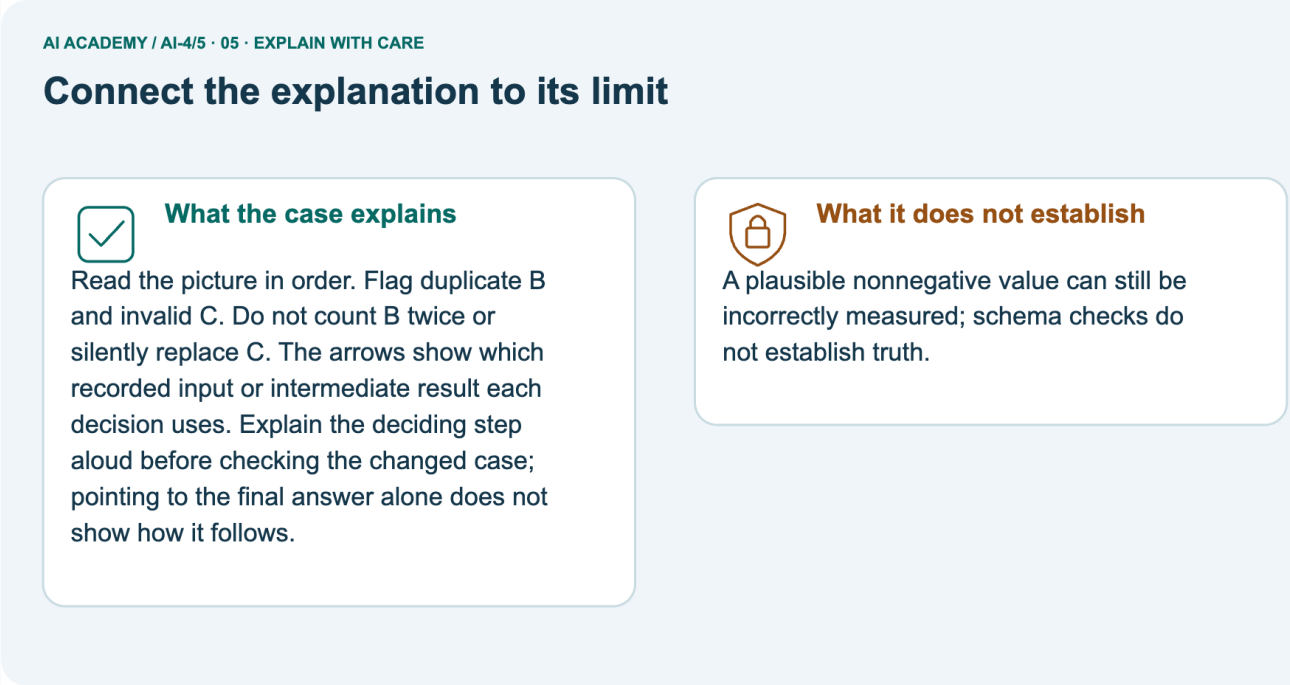
Original case: Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. Changed condition: The teacher confirms B's second row is an accidental **duplicate**.

Expected result: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

> Explain the case, then point to a limit on the claim.

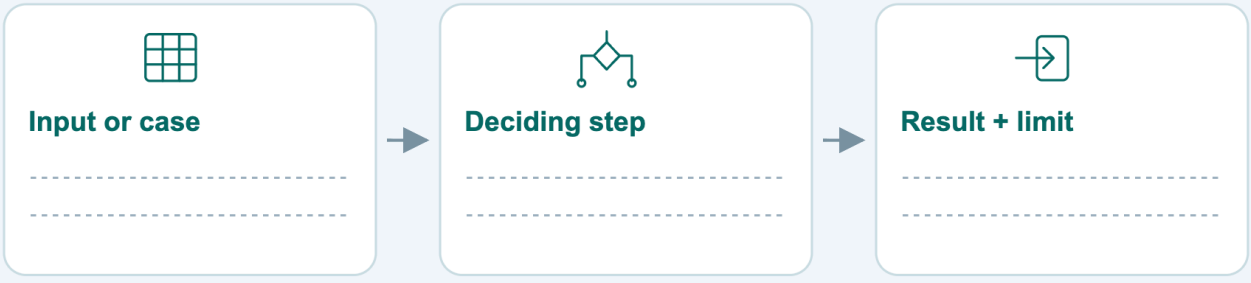
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/5 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.